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Review article



Machine learning in forensic toxicology: Concepts, applications and challenges in bioanalysis, ADME, and toxicodynamics

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ABSTRACT

Forensic toxicology focuses on the detection, quantification, and interpretation of medicinal and recreational drugs, other chemicals or poisons, and their metabolites in biological matrices. Chromatography, combined with mass spectrometry (MS), is the most widely used analytical technique. However, forensic toxicology faces increasing analytical challenges due to a continuously changing drug landscape. In particular, the emergence of new psychoactive substances (NPS) has driven the development of more complex analytical methods (e.g., high-resolution mass spectrometry), novel markers (e.g., metabolomics), or innovative screening approaches (e.g., activity-based), which collectively generate vast amounts of data. These challenges include rapid market dynamics with the constant emergence of new chemical scaffolds and modifications, complex fragmentation and metabolic behavior, and limited or delayed access to reference materials- These developments are not limited to NPS alone. Consequently, machine learning (ML) algorithms have increasingly found their way into forensic toxicology. This review discusses various applications of ML methods related to bioanalysis, metabolomics, and toxicodynamics in the context of forensic toxicology. Currently, a major limitation is the compilation of sufficiently large and suitable datasets, which is often constrained by limited availability of real case data, inhomogeneous analytical data, *in vivo* study designs with small group size (< 10 animals per group), or a low number of included substances. Ultimately, the quality of an ML model relies not only on data quality but also on a thorough understanding of analytical chemistry, biochemistry, pharmacology, medical case history, and ML design, highlighting the importance of interdisciplinary collaboration in these studies.

1. Introduction

Forensic science is the application of scientific principles, techniques, and methodologies to investigate and analyze scientific evidence, such as physical and chemical traces, for use in a legal context [1]. One of these disciplines is forensic toxicology, which focuses on the detection, identification, quantification, and interpretation of the presence of medicinal and recreational drugs, other chemicals or poisons, as

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well as their metabolites in a biological matrix, with the aim to elucidate whether a substance played a (relevant) role in a judicial query [1]. The main methods applied in this discipline are chromatography and mass spectrometry (MS) due to their high sensitivity and specificity. Routine forensic toxicological casework often entails the identification and quantification of a defined set of compounds in biological matrices (e.g., blood, urine, hair etc.) [2–4]. However, research in this field also includes, for example, metabolomics, the pharmacological evaluation of toxicity (structure-activity relationship studies), and the interpretation of post-mortem concentrations in relation to post-mortem redistribution and tolerance development due to an individual's consumption history [5]. As with any other field, the field of forensic toxicology is also continuously evolving, with a particular challenge posed by the emergence of novel compounds on the recreational drug market (e.g., new psychoactive substances; NPS) [6–8]. Other challenges involve understanding the toxicological aspects of overdoses and deaths, the increasing demand for the determination of the cause of death, and the risk of weaponized CBRNE (chemical, biological, radiological, nuclear, and explosives) poisoning. Hence, there is a need for forensic toxicologists to be at the forefront of providing evidence to the judicial systems [9,10]. Therefore, research in this field is constantly needed, and, as will be clear from this review, forensic toxicology can also benefit from recent developments in machine learning (ML).

These algorithms can be used to analyze broad datasets of chemical structures, biological activities, and toxicokinetic parameters, enabling the rapid identification of druggable candidates. By predicting molecular interactions, toxicity, and toxicological ADME (administration, distribution, metabolism, excretion) properties, ML accelerates discovery and reduces the need for costly and time-consuming experimental studies [11]. In supervised ML, learning occurs when an algorithm is given a set of inputs (features) and a known output (also referred to as an endpoint). The paradigm then approximates a learner or predictive function (i.e., the rules) between the inputs and outputs, minimizing a selected loss function, for example, the mean absolute error between the expected and predicted output values [12]. As a result, these rules can be used to predict the properties of unknown substances. In contrast, unsupervised ML aims to identify patterns within a dataset without a predefined endpoint and encompasses a diverse range of techniques, including visualization methods. Accordingly, unsupervised learning is commonly applied for exploratory data analysis purposes (e.g., metabolomics) [13].

To the best of the authors' knowledge, the first application of ML methods in forensic toxicology dates back to 1989, when Spiehler [14] investigated three different paradigms for the interpretation of morphine-involved deaths. The history of quantitative structure-activity relationships in pharmacology can be traced back even further, with examples dating back to the late 1800s and the Meyer-Overton hypothesis [15]. While ML methods have a long history in pharmacology and toxicology, relatively few publications address their application to recreational drugs in forensic toxicology, highlighting a clear gap in the current literature.

This review aims to discuss the current state of the art in the application of ML in forensic toxicology and chemistry and to identify key interdisciplinary research questions for future exploration. Only original research articles presenting newly developed ML approaches were included, excluding studies based on commercially available software.

2. Machine learning methods in pharmacology and analytical chemistry – focus on quantitative structure activity relationship methods (QSAR)

In the context of pharmacology and analytical chemistry, the application of ML methods is driven by the desire to predict, optimize, and address research questions more rapidly compared to traditional laboratory-based experimentation. Beyond enabling research in a cost- and time-efficient manner, ML also enables the investigation and discovery of hidden patterns within complex datasets.

Structure Activity Relationship (SAR) analysis is a computational modeling approach, which establishes a mathematical relationship between the chemical structure of xenobiotics and their biological, chemical or physical properties [16]. The primary aim is to predict the behavior or activity of xenobiotics based on their molecular descriptors. The independent variables are typically molecular descriptors, such as molecular mass or lipophilicity, while target endpoints (dependent variables in classical statistics) determine the mathematical or statistical models that are best suited to derive correlations or draw conclusions. If the endpoint is continuous, quantitative SAR (QSAR) methods are employed, but often the terminology is used interchangeably. This has largely been driven by the vast number of chemicals and medicinal products developed each year that need to be toxicologically evaluated for potential hazardous effects on human health [17].

Most often, chemical structures are encoded as symbolic representations or chemical annotations that computers can easily process. Each descriptor, or attribute derived from a structure, relates to a single property or a combination of different properties of the substance (e.g., lipophilicity metrics such as logD, logP, total polar surface area (TPSA)). Usually, a few thousand descriptors are calculated per substance, using softwares such as Dragon, PaDEL, or MORDRED [18–20]. However, not all descriptors or structural motifs are informative in every use case (application), or have a biologically plausible reason for inclusion. Common endpoints are receptor affinity, receptor activity, solubility, or liquid chromatography retention time (for developing analytical methods). Frequently used algorithms in ML-based QSAR studies include Random Forest (RF), support vector machines (SVM), artificial neural networks (ANN), and other paradigms. Performance is then compared to identify the most robust and informative paradigm that matches the problem space. More advanced methods include deep neural networks and biologically or evolutionary inspired paradigms. To identify the most robust and informative paradigm that matches the problem space, several models are trained and their predictive performance compared. Typical performance metrics are summarized in Tables 1 and 2. Using ML approaches, non-linear, multi-dimensional relationships can be handled, automated feature selection/extraction can be included, and complex datasets can, in general, be handled more efficiently and robustly.

A very important factor to consider when building ML models is the *applicability domain*. This term refers to the (chemical) space for which a model can give reliable predictions [21]. This is determined by the nature of the data used in training. For QSAR studies, predictions are generally limited to chemicals/structures similar to those used in the training set [22,23]. Chemicals within the model's scope are considered to be within the applicability domain and are therefore interpolated; those outside the scope are considered extrapolations, and their predictions are consequently less reliable [24]. The applicability domain of a QSAR model can be expanded by using a chemically and structurally diverse training set. In QSAR studies, the applicability domain is often regulated by guidelines such as those of the OECD (Organization for Economic Co-operation and Development) [25]. The five OECD principles for QSAR validation are prerequisites for any model proposed for regulatory use and are used to demonstrate the validity of a QSAR model [24]. These guidelines formally require an applicability domain definition to ensure regulatory robustness. The interested reader in ML methods in pharmacology and analytical chemistry is referred to the books by Todeschini and Consonni, "Handbook of Molecular Descriptors," [26] as well as by Roy, "Cheminformatics, QSAR and Machine Learning Applications for Novel Drug Development" [27].

3. Bioanalysis

3.1. High-resolution mass spectrometry (HRMS) analysis

Nowadays, the primary tools for the qualitative and quantitative

Table 1
Typical performance metrics for classification models.

Metric	Formula	Explanation	Interpretation
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Probability of correct classification	Value between 0 and 1 with higher values indicating better performance
Sensitivity, Recall, True Positive Rate	$\frac{TP}{TP+FN}$	Probability of a positive result if the instance is positive	Value between 0 and 1 with higher values indicating better performance
Specificity, True Negative Rate	$\frac{TN}{TN+FP}$	Probability of a negative result if the instance is negative	Value between 0 and 1 with higher values indicating better performance
Precision, Positive Predictive Value	$\frac{TP}{TP+FP}$		Value between 0 and 1 with higher values indicating better performance
F1 score	$\frac{2TP}{2TP+FP+FN}$	Harmonic mean of the precision and recall.	Value between 0 and 1 with higher values indicating better performance
MCC (Matthew correlation coefficient)	$\frac{TP \times TN - FP \times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$	Performance evaluation of binary classification	Value between −1 and 1 with higher values indicating better performance

TP: true positive, FN: false negative, TN: true negative, FP: false negative

Table 2
Typical performance metrics for regression models.

Metric	Formula	Explanation	Interpretation
R^2 (coefficient of determination)	$1-\frac{\sum\lvert y_i-\hat{y}_i\rvert^2}{\sum\lvert y_i-\bar{y}_i\rvert^2}$	Percentage of the variance in the dependent variable that the independent variables explain collectively	Value between 0 and 1 with higher values indicating better performance
RMSE (root mean square error), RMSEE (root mean square error of estimation), RMSEP (root mean square error of prediction)	$\frac{1}{N}\sum_{i=0}^{N-1}\lvert y_i-\hat{y}_i\rvert^2$	Measure on how far the predicted values deviates from the measured value using Euclidean distance.	Values closer to 0 indicate better performance
MAE (mean absolute error)	$\frac{1}{N}\sum_{i=0}^{N-1}\lvert y_i-\hat{y}_i\rvert$	Average absolute difference between predicted and measured values	Values closer to 0 indicate better performance
MedAE (median absolute error), MAD (median absolute deviation)	$median(\lvert y_i-\hat{y}_i\rvert)$	Median absolute difference between predicted and measured values	Values closer to 0 indicate better performance
MRE (mean relative error)	$\frac{1}{N}\sum_{i=0}^{N-1}\frac{\lvert y_i-\hat{y}_i\rvert}{\lvert y_i\rvert}$	Percentage deviation of model prediction from ground truth	Values closer to 0 indicate better performance
Q2	$1-\frac{\sum\lvert y_i-\hat{y}_{ext}\rvert^2}{\sum\lvert y_i-\bar{y}_i\rvert^2}$	Measures the model's ability to predict new data	Values closer to 0 indicate better performance

y: dependent variable, ŷ: predicted value of y, ȳ: overall mean of y

analysis of biological specimens, including those for medicinal and other drugs, as well as other chemicals, in clinical and forensic laboratories are methods based on liquid chromatography coupled to single (LC-MS) or tandem mass spectrometry (LC-MS/MS) [28]. Due to the constantly evolving NPS market, with the continuous occurrence of novel compounds, scientists find it challenging to develop and maintain up-to-date targeted screening methods. Hence, non-targeted screening methods using high-resolution mass spectrometry (HRMS/MS) are needed. However, when HRMS/MS methods with data-independent acquisition (DIA) for non-targeted screening are applied, large amounts of data are generated since all ionized analytes in a sample are fragmented and their MS/MS spectra are acquired. Consequently, data evaluation can be time-intensive and elaborate. The advantage of DIA is that retrospective data analysis is possible, as it does not rely on *a priori* defined criteria (e.g., the number and absolute abundance of precursor ions selected for fragmentation), but instead acquires fragment ion data for all precursors within predefined *m/z* windows [16]. The most common approach for analyzing HRMS DIA data is the use of mass spectral databases/libraries. However, this approach is limited by the number of included compounds.

Recently, Streun et al. [29] evaluated two machine learning platforms (KNIME and keras with TensorFlow) [30] and a custom-made ML algorithm for the classification of HRMS DIA data files from solvent and whole blood samples with and without medicinal drugs and drugs of abuse common in forensic toxicology. A future application for these algorithms would be the automatic evaluation of screening data in forensic laboratories. All three tested platforms used ANNs. The visual, node-based open-source analytics tool KNIME does not require programming knowledge, as it is built on a plug-and-play approach that encapsulates different steps of an ML workflow, allowing users to

combine and configure their tasks through a graphical user interface. Compared to this, Keras is a high-level application programming interface (API) that runs on top of TensorFlow, a deep learning framework, where workflows are built in code, offering more flexibility (while requiring extensive programming and ML modeling experience). The custom-built ANN algorithm was developed in Python, enabling full customization and potentially more tailored data preprocessing or modeling techniques. The study demonstrated that the trained ANNs achieved sensitivity and specificity levels suitable for routine laboratory use, with minimal differences observed across the software platforms employed, indicating that the choice of platform does not significantly affect the ANN's performance for the task at hand.

Additionally, Streun et al. [31] developed an ANN to identify manipulated urine samples (an important matrix for abstinence control, workplace drug testing, and doping control) using HRMS IDA screening. The advantage of this approach is that it could be implemented in routine HRMS screening to automatically flag atypical urine samples for further analysis.

Recently, Huang et al. [32] reported on an in-house software tool (MSIE) for automated elucidation of (1) eight NPS classes, (2) six synthetic cannabinoid receptor agonist (SCRA) subclasses, and (3) chemical structures of unknown SCRAs, based on nontargeted HRMS/MS collision-induced dissociation (CID) or electron-activated dissociation (EAD) data taken with an ExionL UHPLC coupled to a ZenoTOF 7600 (AB Sciex, Framingham, USA). The software, coded in MATLAB with a Python environment, was trained on a CID dataset of 505 NPS (183 SCRAs, 95 synthetic cathinones, 46 phenethylamines, 48 fentanyl analogs, 14 tryptamines, 22 benzodiazepines, 14 phencyclidines, 6 piperazines, 2 aminoindanes, 3 metabolites, 3 stimulants and 69 other medicinal and recreational drugs) and the EAD dataset of 181 SCRAs. Of these, 55 (CID dataset) and 31 (EAD dataset) were reserved for model validation. Four classifiers were evaluated separately for each task: k-Nearest Neighbors (kNN), SVM, Extra Trees (ET), and soft voting ensembles (kNN, SVM, and ET). Model optimization was performed using a Bayesian optimization algorithm and nested cross-validation. The parameters used for model evaluation were precision, recall, and F1 score. For (1) and (2), voting ensemble models achieved the highest F1 scores: 93.3 % for the NPS classification model (CID dataset), 90.6 % for SCRA subclass classification (CID dataset), and 95.3 % for SCRA subclass classification (EAD dataset). The models for structure elucidation were only developed for SCRAs, whose structures were divided into four groups: head, linker, core, and tail. Of 181 SCRAs, 36 were not correctly recognized (33 due to missing fragment information and three due to unusual structures). The MSIE software was tested on 39 real case samples. The NPS nontargeted screening model classified the samples with an accuracy of 93 %, and a subsequent SCRA classification model classified 16 samples containing 39 SCRAs with an accuracy of 95 %. All 39 SCRA structures were successfully elucidated. A key limitation of the study was that the subclass classification and structure elucidation were restricted to SCRAs. Furthermore, the superior performance of the EAD model was tied to the ZenoTOF 7600 platform, which may limit the broader applicability of MSIE in laboratories without this specific instrumentation. Last, as was also evident from the paper, the methodology may still struggle with unusual (unanticipated) structures.

3.1.1. Retention time and fragmentation predictions in LC-HRMS(/MS)

When it comes to the retrospective analysis of DIA LC-HRMS/MS data, ML can be a helpful tool for predicting LC retention times of novel compounds such as NPS, metabolites, or novel performance enhancers [33,34], particularly where reference standards are not available. Various ML and statistical methods have been evaluated for retention time prediction across fields such as environmental and analytical chemistry [35,36]. For example, linear regression using log-K_{ow} as the only predictor was successfully applied for the retention time prediction of contaminants in wastewater (with a window of ± 2 min for 70 % of compounds) [35], and combinations of kNN clustering with genetic algorithms (GA) and SVMs were applied for the retention time prediction of contaminants in environmental samples [36].

Li et al. [37] presented a novel approach for LC retention time prediction, called multiple datasets learn-transfer learning (MDL-TL), which utilizes multiple datasets (n = 14). This approach encodes the chromatographic column information and operational conditions, pretrains on the union of multiple datasets, and transfers the learned knowledge to the target task (retention time prediction). Hence, it can be applied to HRMS data independent of the LC chromatographic system. The MDL-TL model was compared with five published retention time deep learning approaches: 1D convolutional neural networks (CNN) [38], graphic neural networks (GCN) [39], relational graph convolution network (RGCN) [39], graph neural network- retention time (GNN-RT) [40] and graph isomorphism network (GIN) [41] and outperformed them in terms of their R [2], mean absolute error (MAE, average of the absolute values of residuals), median absolute error (MedAE), and mean relative error (MRE) on almost all datasets. However, a correlation was observed between the similarity of the chromatographic conditions of the datasets and the performance of MDL-TL (i.e., less similarity, worse performance). While the applicability of MDL-TL was shown using data from the OSI database (an in-house database by Zhao et al. [42]) in combination with candidate test analytes from the human metabolome database (HMDB), with a high identification accuracy and the reduction of false positives, its application on real (forensic) samples still needs to be proven.

Miller et al. [33] tested different ANN designs (linear, radial basis function (RBF), probabilistic neural networks (PNN), and multilayer perceptrons (MLPs)) for retention time predictions in archived urine HRMS data from the London 2012 Olympic and Paralympic Games related to performance-enhancing agents. Retention times could be predicted within a time window of ±0.5 min for 93 % of the compounds. To the best of our knowledge, this was the first application of ANN models to predict LC retention times in forensic toxicology.

This approach was further investigated and improved by Pasin et al. [43] who developed a quantitative structure-retention relationship (QSRR)-based MLP model for predicting retention times, optimized for a range of LC-MS systems and tested on NPS. The model utilized four molecular descriptors (logD, logP, number of carbons, number of oxygen atoms), drug classes (n = 13), and laboratory sites involved (n = 9) to account for differences between LC systems. The model achieved an R^2 of 0.942 and a MAE of 0.583 min for the test set, indicating high predictive accuracy. Unlike previous LC retention time prediction models, this novel QSRR model can also predict LC retention times for novel compounds.

Polettini et al. [16] investigated whether ML could be used to identify SCRAs from DIA data obtained with HRMS/MS using molecular formula and retention time information. A total of 105 SCRAs were included, which were measured using an LC-HRMS system, and their retention time was recorded relative to the internal standard JWH-018 N-pentanoic acid-d5. The relative retention time (rRT) was calculated as the analyte's retention time divided by the internal standard's retention time. To build a multiple linear regression model able to predict retention times the 1D, 2D molecular descriptors and fingerprints of the included SCRAs were calculated using PaDEL-Descriptor. Five compounds were excluded from the model evaluation. Feature selection included excluding highly correlated molecular descriptors and applying the Boruta algorithm [44], an ML-driven paradigm. For model validation, repeated k-fold cross-validation was performed, and an external validation using the five excluded SCRAs. The R^2 of the model was 0.8593, the root mean square error was 0.087 (ca. 0.56 min), and the MAE was 0.060. For the five additional compounds used for validation, the absolute rRT between predicted and experimental rRT was within 0.073 (ca. 0.5 min retention time).

In addition to predicting RTs within the context of NPS, predicting MS/MS spectra from a given (known) chemical structure for confirmational purposes in HRMS has been developed. Software such as MetFrag

[45] developed for the identification of novel metabolites, uses heuristic or combinatorial fragment modeling techniques to predict the positions and likelihoods of bond breakages in a molecule and the resulting MS/MS spectra [46]. Another commonly used software for HRMS data evaluation, especially for metabolomics data, is the freeware XCMS, whose workflow consists of feature detection, correspondence determination, and additional context-dependent analysis [47]. Since its initial launch in 2006, various optimization algorithms have been evaluated and implemented, including isotopologue parameter (IPO) optimization and autotuner estimation, which utilize the predictive power of ML as a proxy for dataset quality [48].

3.2. Alternative screening methods

For the identification of novel NPS, HRMS/MS-DIA-based non-targeted screening is now considered the state-of-the-art [49,50]. However, the need for reference spectral libraries, as well as the fact that acquisition is triggered only when a certain threshold is reached (which poses a problem for low-concentration NPS), pose limitations for this method. An alternative are activity-based bioassays, which have the advantage that no knowledge of the structure is required at any point [51]. Furthermore, active metabolites can be detected, addressing the challenge that, early after their emergence, NPS metabolites may not be known, and, hence, may be missed. These assays are intended as a first-line screening, so that only samples containing an active substrate (e.g., an agonist at the μ-opioid (MOR) [52] receptor [52] will be further analyzed using MS confirmatory methods. These assays have been applied not only to biological matrices but also to the screening of plant materials [53] or waxy or putty-like drug preparations containing SCRAs [8]. While most applications target the CB_1 receptor, screening assays have also been developed for e.g., the MOR [54]. It is worth noting, however, that not every NPS substance class is suitable for activity-based screening; for example, the endogenous serotonin baseline signal in body fluids is too pronounced for screening purposes [55]. Typically, these assays are visually and independently scored by at least two experts using a standardized evaluation protocol for guidance. However, as also discussed further (see Section 5.2), ML has also been implemented for automated scoring.

4. Toxicokinetics (ADME)

4.1. Time since consumption

In a clinical setting, to adequately treat patients, it can be crucial to identify which drugs were consumed and the time since consumption. The latter may also be relevant in, e.g., criminal cases. Heroin can be used as a case example for this. Due to the fast metabolism of heroin and its first metabolite, 6-monoacetylmorphine (6MAM), these are often not detected in biological samples. Furthermore, morphine and its metabolites morphine-3-glucuronide and morphine-6-glucuronide are not distinct biomarkers for heroin consumption but can also result from the consumption of codeine, morphine itself, and other morphine derivatives [56]. Alternatively, opiate alkaloid impurities resulting from heroin production have been used to confirm consumption in biological samples. Based on data collected via LC-MS/MS, Dubois et al. [57] developed a model to predict the time since heroin consumption in chronic users. Eleven patients were included in the study, and seven or more plasma samples were analyzed, with the first sample collected between 1.1 and 44.2 h after the last consumption (timing reported by the patients). Additionally, samples from seven test patients (outpatients treated with methadone) were collected. A general linear mixed model (GLMM) approach was applied. The model was assessed using the jackknife technique, that is, by leaving one patient out of the dataset and re-evaluating. The model was then applied to the independent test group. Depending on the analyte, the correlation between declared (patient-reported last consumption time) and predicted delays ranged from 0.88 (morphine-6-glucuronide) to 0.94 (morphine-3-glucuronide). The overall prediction error was about 8–9 h. On the test set, the model performed well in estimating the time since consumption. Although the study design appeared well-structured, it suffers from major limitations. The very small sample size ($n = 11$) severely restricts statistical validity and generalizability. Reliance on self-reported consumption times introduces uncertainty, and the failure to account for dose or to consider multiple intoxications further undermines the model's reliability and interpretability.

4.2. Metabolomics with ML

Metabolomics involves the screening of a diverse set of small-molecular-weight substances, such as metabolites, in a biological system. In metabolomics, the biological system is referred to as the metabolome, which comprises endogenous and exogenous metabolites [58,59]. It involves screening and identifying suitable biomarkers (metabolites) for the respective research question [60]. Both targeted and untargeted approaches have been used for metabolomics studies in combination with ML. The generation of vast amounts of data by (untargeted) HRMS makes it an ideal starting point for applying ML models to data analysis [61].

4.2.1. Metabolomics for screening purposes

Olesti et al. [62] applied targeted LC-MS/MS in combination with ML to investigate metabolome changes of monoamine neurotransmitters and steroid hormones ($n = 18$) in rats after the administration of different recreational drugs (MDMA, methamphetamine, cocaine, heroin, THC, mephedrone, and JWH-018). Plasma, urine, and brain tissues (hippocampus, prefrontal cortex, cerebellum, striatum) from rats ($n = 6$ per drug, $n = 6$ per vehicle) were analyzed at two time points (1 and 4 h post application), using two different LC-MS/MS methods. First, ratios between drug-treated and vehicle-treated groups for each metabolite were calculated, and data imputation was performed using a regularized iterative principal component analysis (PCA) algorithm. Data normality was achieved by log-transformation, which was confirmed using the Shapiro-Wilk test. Sample clustering was performed by drug class using PCA, and Mann-Whitney U-tests were used to detect significant metabolite changes with 95 % confidence. An RF classifier was trained on the metabolomic fingerprints and evaluated using Leave-One-Out Cross--validation (LOO-CV), achieving 100 % accuracy on classic recreational drugs. Further variable importance analysis identified serotonin, dopamine, and norepinephrine as the most critical pathways. Apart from LOO-CV, the possibility of overfitting was mitigated by building 168 smaller models, with progressively added features. The approach itself is interesting, although it requires animal experiments, as brain tissue metabolites were more predictive than plasma or urine markers, which consequently poses the problem of interspecies differences. In addition, while the authors claim that their approach is a fast and cost-effective pharmacological tool to identify NPS according to metabolomic changes, there have been several other approaches, such as the application of *in vitro* live-cell-based screening methods to show that a drug targets a specific receptor and, hence, behaves comparably to other recreational drugs [8,63,64].

Recently, Streun et al. [65] presented a new qualitative screening assay for SCRAs based on metabolomics data and ML. Essentially, the RF model was trained to distinguish between SCRA-positive and SCRA-negative urine samples based solely on global metabolomic changes (e.g., alterations in lipid metabolism). Hence, the model does not rely on the knowledge of specific SCRAs or their metabolites for classification. It assumes that SCRAs, due to shared pharmacological targets (CB_1 and CB_2), influence similar downstream processes, thereby leading to similar metabolic changes. The study included 474 urine samples, of which 201 had previously tested positive for at least one SCRA (25 different SCRAs in total). Samples were analyzed with untargeted HRMS, and raw data were pre-processed with Progenesis QI

2.4 (Waters Corporation, MA, United States). Based on the identified predictive features, an RF model was trained using 10-fold CV with a balanced training set (n = 176 positive and 193 negative samples). Additionally, a prediction analysis was performed using the ROC thresholding method with the selected features, and as an unsupervised control, PCA and k-means clustering with two clusters were used. The optimized model achieved a mean accuracy of 88.1 %, a clinical sensitivity of 83.0 %, a clinical specificity of 92.7 %, a positive predictive value of 91.3 %, and a negative predictive value of 85.6 % using the training set. The lower clinical sensitivity relative to specificity suggests that some positive samples may be misclassified as negative. The test set prediction yielded a mean accuracy of 88.0 % (positive samples: 80.0 %, negative samples: 96.0 %). The model can serve as a universal screening tool for unknown SCRAs if they cause similar metabolic changes. However, this poses two problems: first, the model cannot distinguish between phytocannabinoids (THC, CBD, etc.) and SCRAs, and second, it may not classify a new SCRA correctly if it has a significantly different metabolic effect (applicability domain).

4.2.2. Metabolomics in post-mortem investigations

In other forensic contexts, HRMS and ML have also been used. It is hypothesized that at the time of death, the metabolome reflects the events leading up to death and potentially also discloses the cause of death [66–70]. Hence, post-mortem metabolomics may offer the potential to explore a diverse set of research questions in death investigations. To evaluate and investigate the robustness and applicability of post-mortem metabolomics, Elmsjö et al. [69] built an ML model to identify pneumonia as the cause of death. Pneumonia was chosen as there is a well-established procedure for its determination as cause of death, and it is expected that it shows significant effects on the metabolome. The training set consisted of 54 pneumonia cases and 54 matched control cases, and the validation set consisted of 122 pneumonia cases and 174 random control cases. Femoral blood samples from these cases were analyzed using LC-HRMS and processed using multivariate analysis to identify post-mortem biomarkers. Even though the approach of matched control cases limited the number of included cases in the training set, the unsupervised PCA with 1356 features revealed no clear outliers or unwanted trends with regards to gender, age or BMI. Furthermore, supervised PLS models based on gender, age, BMI, or acquisition batch showed no or only limited trends and hence a low risk of confounding. A supervised orthogonal partial least-squares discriminant analysis (OPLS-DA) model, with variable importance in the protection plots (VIP) for interpretation, was built to identify variables (metabolites) contributing to group classification between control and pneumonia cases. These metabolites were identified using an in-house database, the HMDB [71], and the METLIN database [72], by matching molecular weight and retention time. This is a classification model commonly used in metabolomics because it enhances group separation by removing orthogonal (unrelated) variation and is suitable for (relatively) small datasets, such as metabolomic datasets, which often have many features (metabolites) but relatively few samples. Although the Q2 value (the model's predictive ability; Q2 > 0.5 indicates good predictive ability) of 0.44 in CV suggested moderate predictive power, external validation on the validation set yielded a sensitivity of 86 % and a specificity of 84 %. Furthermore, several of the identified biomarkers have also been found in ante-mortem samples from pneumonia patients.

Building upon the results from the study by Elmsjö et al. [69] , Ward et al. [70] from the same group went a step further and trained classification models for screening purposes to differentiate metabolomics data from more than 4000 different post-mortem cases into five different cause-of-death (CoD) classes, in addition to pneumonia: acidosis, drug intoxication, hanging, and ischemic heart disease. After LC-HRMS analysis, the metabolites were identified by matching their mass-to-charge (m/z) ratios to public databases such as the HMDB and METLIN. Feature (metabolite) selection was performed using a variable-influence-on-projection threshold of > 1 to remove less significant metabolites and thereby improve model performance. A one-way ANOVA revealed that 31 metabolites exhibited significant differences in abundance across the CoD groups. A hierarchical clustering was performed to identify distinct metabolic patterns for each CoD group. Initial unsupervised PCA for each CoD group identified fewer than 3 % of cases as potential outliers (n = 119). These were excluded from the dataset. To control for confounders such as age, sex, body mass index, and post-mortem interval, PLS and PLS-DA were applied. The primary classification model was OPLS-DA, which was cross-validated with a 7-fold partitioning method. The initial OPLS-DA model had an R^2 of 0.33 and a Q2 of 0.21, indicating low performance. Hence, feature selection was performed, and the model performance improved to an R^2 of 0.54 and a Q2 of 0.30. Furthermore, the training set had a true-positive prediction rate of 68 %, and the test set had a rate of 65 %. The OPLS-DA model performed best in predicting acidosis (98.7 % accuracy) and pneumonia (93.6 % accuracy), whereas ischemic heart disease had the lowest accuracy (73.1 %).

In medico-legal investigations, an accurate estimate of the post-mortem interval (PMI) provides critical information. This task is commonly conducted by forensic physicians using Henssge's nomogram in conjunction with the three classical signs of death (algor mortis- decrease of body temperature, livor mortis- pink-purple discoloration of the skin where blood pools due to gravity, and rigor mortis- stiffness of muscles). Still, this estimate can be greatly influenced by ambient temperature, weather, wind [73] and the forensic physician [74]. The degree of decomposition also significantly influences the robustness of the estimation [75]. Recently, Løber et al. [76] developed ML models to estimate PMI (4-day interval) based on ante-mortem metabolome modifications using HRMS data from various rat tissues (brain, vitreous humor, blood, and muscle). Rats (n = 50) were euthanized using carbon monoxide on the same day and divided into 5 groups (n = 10), each corresponding to a sampling day (0, 1, 2, 3, 4 days after death). An additional two rats were euthanized using pentobarbital to evaluate the potential impact of the cause of death on post-mortem metabolomic changes. Between death and sampling, rats were placed on a steel table in a prone position at a room temperature of 19.5–21 °C. The collected tissues were analyzed using HRMS. For model development, a nested LOO-CV approach was used: one mouse was held out for testing, and one was used to estimate model robustness. Following data preprocessing, feature selection was performed using the least absolute shrinkage and selection operator (LASSO) and RF algorithms, and next, LASSO and tree-based models were constructed from these features (biomarkers, n = 15). Model performances were assessed via the bootstrapped model performance parameters: R^2, RMSE, and confidence interval. The final models were validated using five rats from an independent validation study conducted 13 months later. Muscle tissue had the highest overall R^2, followed by brain and blood. Generally, the RF model performed better than the LASSO model, which is supported by the poor performance of LASSO (R^2 ranging from 0.622 to 0.934, accuracy ranging from 12.34 to 35.42 h) in the validation, as opposed to the RF models (R^2 ranging from 0.925 to 0.984, accuracy ranging from 3.66 to 7.84 h). No difference in metabolic changes between the two euthanization methods was observed; however, the sample sizes of 50 vs. 2 are highly imbalanced. While this study was conducted in rats, all annotated metabolites, except for anserine, are common between humans and mammals. Future studies are needed to assess whether the findings may be extrapolated to human subjects.

4.2.3. Metabolomics in the context of medicinal and recreational drugs

Gamma-hydroxybutyrate (GHB) is an endogenous short-chain organic acid [77], a prescription medication [78], but also consumed as a recreational drug or abused in relation to drug-facilitated crimes or sexual assaults [79]. It is rapidly metabolized in the body and therefore has a narrow detection window of up to 6 h in blood and 12 h in urine [80], which renders the detection of exogenous GHB in forensic toxicological casework difficult [81]. Wang et al. [82] investigated

metabolomics to identify suitable GHB biomarkers, thereby confirming GHB consumption. The dataset consisted of one group of whole-blood analyses from 51 drivers under the influence of GHB (n = 51) and a second matched control group (n = 51). Following HRMS data processing and normalization, a pairwise univariate *t*-test was performed to assess the difference for each feature between groups (positive vs. control) using false discovery rate (FDR) to account for multiple testing, and PCA for multivariate analysis. Pearson correlation followed by correlation network analysis (CN) was performed to identify potential metabolites and to understand the impact of GHB on the metabolism. Feature selection was performed using OPLS-DA with Pareto scaling and validated with 10-fold cross-validation. Models based on LASSO and RF (random RFF and classification RFC) were used to predict metabolites associated with exogenous GHB intake. Evaluation showed that the OPLS-DA model (R2Y = 0.86, R2X = 0.42, Q2 = 0.36, and the returned root-mean-square error of estimation (RMSEE) is 0.19) provided more information for metabolite identification than the LASSO (R^2 = 0.96, RMSE = 0.27) and RF models (RFR: accuracy = 0.89, RFC: RMSE = 0.62, R^2 = 0.80).

Fatal intoxications are not always caused by the consumption of recreational drugs or poisons but can also result from toxic concentrations of medicinal drugs. However, the interpretation of post-mortem samples is difficult due to post-mortem redistribution [83] and the overlapping ranges of toxic and fatal concentrations [84], especially in instances where tolerance may occur [85]. In this context, Ward et al. [86] explored the utility of metabolomics for determining hypoglycemia-related deaths, including insulin intoxication, using a multivariate model. While hypoglycemia-related death can occur due to a multitude of health reasons (e.g., malnutrition, prolonged fasting, or organ failure) [87], in the forensic context, synthetic insulin intoxications from accidents, homicide, and suicides are of particular interest [88]. While physiological insulin concentrations are crucial for stabilizing blood glucose levels [89,90], high concentrations can cause hypoglycemia and ultimately death. In forensic investigations, the post-mortem diagnosis of insulin intoxication is challenging due to the absence of pathognomonic findings at the autopsy and the fast metabolism of insulin [91]. Three different matched study groups (1:1:2, matched for age, sex, and body mass index (BMI)) for building the model were included: confirmed insulin overdose (group name "hypo", n = 19), hyperglycemic deaths (group name "hyper", n = 19), and a control group (cause of death: hanging, n = 38). The model was tested on a random test group (n = 776) for which screening data were available. Following peak processing and data normalization of the HRMS screening data, multivariate supervised OPLS-DA models were constructed and demonstrated good performance (hypo vs. control OPLS-DA model: R^2 = 0.96 and Q2 = 0.72, with a CV-ANOVA p-value < 0.001; hypo vs. hyper OPLS-DA model: R^2 = 0.96 and Q2 = 0.86, with a CV-ANOVA p-value < 0.001). The combination of these two OPLS-DA models into a SUS plot identified 85 discriminatory features for the hypo group, and the consecutive three-group OPLS-DA model based on only these 85 features could differentiate the groups (R^2 = 0.76, Q2 = 0.46, CV-ANOVA < 0.001). Screening of the test set classified 46 cases as hypo, 26 as hyper, and 322 as control, with 378 cases not exceeding the probability threshold of 0.65 required for above-random class prediction. Although the model performed well, the sample size of the three groups is a limitation. The group around Ward and Elmsjö et al. used the same methodology as in their insulin study [86] to further investigate post-mortem metabolomics of hypothermia [92] and fatal oxycodone-related intoxications, revealing acylcarnitines as potential markers for fatal opioid intoxications [93].

Similarly, Dai et al. [94] investigated the application of metabolomics to identify endogenous biomarkers for fatal estazolam (tranquilizer, benzodiazepine) intoxications using a mouse model. Overall, the aim was to build a classification model that could differentiate estazolam fatal intoxications from other causes of death. Two groups of mice were administered a lethal dose of estazolam (n = 10, 6 × LD_{50}) or a poisonous dose of estazolam (n = 10, 50 × and 100 × therapeutic doses), respectively. The study used five different control groups (n = 10), including mice that were killed by cervical dislocation, drowning, mechanical asphyxia, or acute hemorrhagic shock, and an additional control group with intoxications of other sedative-hypnotic drugs (e.g., diazepam, zaleplon, nitrazepam, and sodium pentobarbital). Collected plasma and brainstem tissue samples were analyzed using LC-HRMS/MS. For class separation and feature importance, a PLS-DA model was built, which showed good predictive performance (plasma: R^2 = 0.99, Q2 = 0.90; brainstem: R^2 = 0.99, Q2 = 0.86; 100 permutation Test: p < 0.01). Two SVM models (one for plasma, one for brainstem tissue) based on multivariate ROC analysis were built for classification, validated using 10-fold cross-validation and 100-permutation testing. These had a high accuracy (plasma: AUC = 1.0, brainstem: AUC = 0.996), also after 20 days at 4 °C (AUC = 0.971). Although the model was extensively validated, the observed high AUC values may suggest potential overfitting, which is especially relevant given the high-dimensional nature of metabolomics data and the relatively small sample size.

Wang et al. [95] also used an animal model and ML to predict the time interval since ketamine administration in rats, based on metabolomics. To this end, these authors developed an algorithm to process HRMS data obtained from samples collected at 21 time points (0 h to 29 days) after ketamine administration (0, 30, 60, and 90 mg/kg) in 35 rats. The data were first processed with Profinder for automatic peak picking, and only peaks detected in 50 % of samples in a test group were used for metabolomic data processing to identify suitable biomarkers. This consisted of multivariate analysis, including PCA, PLS-DA, and ANOVA. Thirty-nine time-dependent potential biomarkers were identified and used for model development. An RF model was trained to estimate time since administration and evaluated using 10-fold cross-validation (9:1). The model achieved 85.4 % accuracy on the CV set, with an average area under the ROC curve (AUROC) of 0.96. A recall of 0.85 indicated high sensitivity. The high Matthew correlation coefficient (MCC, a metric to evaluate the performance of binary classification, with a range of −1 (poor prediction) to + 1 (perfect prediction)) of 0.81 indicated that the predicted model was reliable. However, on the validation set (24 samples), accuracy was only 58.3 %. The authors attributed this to sampling time points differing between the test and validation sets.

4.2.4. Metabolomics and time of deposition

The mere presence of a trace does not automatically imply that it is related to (e.g., was deposited at the time of) the offense. Crime scene investigation methods are needed to estimate the time since deposition of a trace. Schneider et al. [96] utilized metabolomics combined with ML to identify biomarkers for estimating the time since blood traces were deposited. Venous blood from eleven volunteers was stored as dried blood spots (DBS) for 0.5, 1, 3, 7, 14, 28, 56, 84, 168, and 334 days, either under controlled laboratory conditions or outdoors in a cage protected from direct physical impact. Analysis of the samples was carried out using LC-HRMS/MS, and raw data were processed using Progenesis Qi. Roughly 19,200 metabolites were identified via LC-HRMS/MS, of which 4700 were allocated to the blank samples, and 11,500 remained after further filtering. Hierarchical clustering, PCA, PLS-DA, and Spearman-ranked correlation analysis further narrowed down the list of candidate metabolites (phenylalanine, indole-3-acrylic acid, spermidine, and argininosuccinic acid). Samples were separated by ageing time and storage conditions using PCA and PLS-DA, and these analyses showed good clustering by time since deposition, especially for samples older than 28 days. It was further shown that storage conditions significantly impacted the metabolome, and therefore separate models were needed for each storage condition. Linear regression models were fitted for each metabolite, showing a strong linear or logarithmic relationship with time since deposition. It was further demonstrated that phenylalanine had the best overall performance with R^2 of 0.99 and 0.93 under laboratory and outdoor conditions, respectively. Spermidine and

indole-3-acrylic acid performed moderately but were less stable over different conditions. Jackknife resampling was performed for internal model validation and to estimate the accuracy of time-since-deposition predictions. The median prediction errors were typically within + /- 15–20 % for longer time since deposition (83–344 days), and a higher variability and lower accuracy were observed for very early time points (0.5–3 days).

5. Toxicodynamics

In medicinal chemistry, the binding affinity and receptor activity of a lead compound are important measures for drug development [97]. While similar methods can be applied in forensic toxicology, their purposes differ. In forensic toxicology, most receptors of interest are those that are associated with psychoactive effects (e.g., CB_1, as opposed to CB_2) [98] and/or adverse health effects (e.g., MOR, which is responsible for respiratory depression after opioid overdose). Hence, in forensic toxicology, the study of the SAR of recreational drugs is used to evaluate their relative harmfulness and to provide recommendations on whether a substance should be prioritized for legal regulation. As *in vitro* SAR studies of recreational drugs are time-consuming and expensive, research has been performed to utilize ML methods to predict toxicodynamic parameters. The following section provides an overview of the application of ML methods in SAR studies, focusing on forensically relevant receptors.

5.1. Structure activity relationships

One useful parameter in SAR studies for assessing the harmfulness of NPS is agonist receptor binding affinity (Ki), which quantifies how strongly a xenobiotic binds to a receptor and, for agonists, must exceed a certain threshold to enable receptor activation. Based on 1170 compounds compiled from ChEMBL [99] (CB_1 agonists, K_i <10000 nM), Paulke et al. [100] developed a predictive model for CB_1 affinity of cannabinoids. For each compound, 210 chemically advanced template search (CATS) descriptors were calculated. The quantitative affinity of an unknown substance was predicted by identifying sufficiently similar neighbors (feature-pair distribution similarity, cutoff: 0.77) and using those to interpolate the value, while accounting for the weighted distance in the training set (inverse distance weighting). The model was solely evaluated using the coefficient of linear correlation after log transformation of the affinity (10-fold cross-validation, $R^2 = 0.78$). The test set consisted of 16 SCRAs for which experimental K_i data had not been previously provided to the model, with predicted K_i values ranging from 20 to 468 nM. The K_i value of the main psychoactive component of *Cannabis sativa*, tetrahydrocannabinol (THC), was correctly predicted at 40 nM [101]. However, it was not reported whether the training set was screened for any compounds of the test set, and dronabinol, a THC-based stimulant and antiemetic, is found in the ChEMBL database [102] (entry ChEMBL465).

In a similar study, Ciallella et al. [103] built models to predict the receptor binding affinities of benzodiazepines at the $GABA_A$ receptor using data from the public databases ChEMBL, BindingDB [104] and the PDSP K_i database [105]. The curated dataset included *in vitro* K_i data of 255 compounds (duplicates were excluded and the lowest K_i retained). Two types of molecular fingerprints (extended-connectivity fingerprints and functional-connectivity fingerprints) and physicochemical descriptors from RDKit [106] were calculated. The applied models were built using RF and SVM. Five models were built in total, with 200 randomly selected molecules, and evaluated using five-fold CV with an 80 % training and 20 % external validation split. Cross-validation showed an internal R^2 of 0.703 and an external R^2 of 0.708. Overall, the predictions were reliable for molecules with $\geq$ 65 % similarity to those in the training set.

Lee et al. [107] used a PLS regression model to predict the CB_1 affinity (pK_i) of SCRAs using molecular descriptors. The measured binding affinities within the dataset ranged from 22 nM (AM-694) to 11,995 nM (RCS-4; CP47,497-C6). The model performed robustly, as indicated by an R^2 of 0.864 and successful validation through y-randomization. However, to ensure consistency, they focused exclusively on substances for which binding affinity was measured within their own laboratory, resulting in a dataset of only 15 substances (eleven for training, three for testing, and one excluded as an outlier). This restricted applicability domain was evident during external validation on 62 naphthoylindole cannabinoids (JWH compounds), which negatively affected predictive performance ($R^2 = 0.70$).

5.2. Toxicological effect

In a study to investigate *in silico* methods as a tool for harm reduction, Rodrigues and Bruni [108] utilized 21 amphetamines paired with 21 cathinones (based on homologous structures). The molecular descriptors were calculated using five different commercial or openly available software tools. Feature selection was performed by multivariate PCA and expert forensic toxicological opinion. A PLS-DA model was built to differentiate amphetamines from cathinones (based on the fact that amphetamines have higher logP values) with an R^2 of 0.91 and a Q2 of 0.83. The prediction of IC_{50} (a toxicological measure of cell viability inhibition) was performed with a PLS regression model with the following performance indicators: R^2 of 0.88, Q2 of 0.77, RMSEE of 6.9, and an RMSEP (Root Mean Square Error of Prediction) of 7.4. While the model was sound and robust, the data size was very small, and the IC_{50} as a representative value for toxicity was questionable. Most experimental IC_{50} data are from either *in vivo* animal studies or from *in vitro* cell assays and cannot easily be extrapolated to humans (applicability domain). Further, the term IC_{50} is not a unique identifier, since it is used for various types of assays, ranging from cell toxicity studies to receptor binding assays.

Recently, Janssens et al. [49]presented an RF model for the automated classification of serum samples in a large-scale evaluation of activity-based screening data for SCRAs. The model was cross-validated against two experts manually scoring the results. The study included 968 blind-coded serum samples, which were analyzed using a CB_1 live-cell-based nanoluciferase assay and HRMS-IDA for confirmation purposes. Two different numerical feature representations were extracted from the data: Tsfresh [109] features and polynomial fit coefficients. The experts had a sensitivity of 94.6 % and a specificity of 98.5 %, and the RF model had a sensitivity and specificity of 94 % (when the prediction threshold was set to 0.055).

6. Discussion and outlook

Forensic science is a multidisciplinary field, and although forensic toxicology is a specialization, research in this area needs to address a multitude of forensic and judicial problems. While the first application of ML to forensic toxicology dates back to 1989 [14], it remains a relatively novel approach in this field, as evidenced by available research and conference abstracts. The majority of ML applications in forensic toxicology are related to HRMS (e.g., retention time prediction, MS data evaluation) [16,110] and metabolomics [69,96]. While HRMS-based ML approaches provide clear practical benefits for screening, data processing, and interpretation, their application to NPS analysis remains associated with several challenges. These include limited availability of representative NPS datasets, instrument- and method-specific dependencies, and insufficient assessment of cross-instrument generalizability.

Although ML models for predictive toxicology have been widely applied in medicinal chemistry and pharmaceutical research, we found only a few publications concerning recreational drugs. One reason could be that the focus of forensic toxicological and medicinal chemistry SAR research is different. For forensic toxicologists, the aim is to identify harmful compounds, to aid in the interpretation of intoxications, but

also, as a direct consequence, to provide an objective ground why legal control may be relevant – or not. Of note, the interests of the forensic toxicology field may intrinsically be somewhat distinct from those of the medicinal chemistry field. In essence, in medicinal chemistry, the aim is to identify compounds that may be beneficial for patient treatment, with minimal adverse effects. However, history has learned that for the forensic toxicologist, compounds that show a less favorable medicinal profile (e.g., because of effects on the central nervous system) may exactly be the ones that are desired from the point of view of a recreational drug supplier or user. Examples include SCRAs [6], opioids [111, 112], and psychedelics [113]. Hence, a model built for medicinal chemistry or pharmaceutical purposes may not be directly applicable to address a forensic toxicological question, as the applicability domain of these two fields differs. However, the underlying ML methods can still be evaluated for their suitability for addressing specific forensic toxicological research questions, such as retention time prediction, support for non-targeted screening, or evaluation of metabolomics data, provided that an appropriate dataset is available. Advances made in one field could and should therefore be evaluated and adapted to enhance the other.

Often, the application of ML in forensic toxicology focuses on simplifying processes or generating and interpreting toxicological data; hence, there is a need to develop ML approaches that can be implemented in routine forensic laboratories and therefore be used in casework. However, for the application of ML in forensic toxicology it is beneficial – and possibly even crucial – that these models are explainable and provide both transparency and traceability. As such, it can be justified to use ANN models, which are not as easily interpreted as, say, rule-based learners like decision tree induction, but have been shown to be highly performant for screening purposes (retention time prediction models) and hence, for assisting in the detection of unknown compounds. However, the models need to be developed on an adequately sized and curated (applicability domain) dataset, and the robustness of the models needs to be sufficiently shown. When it comes to the application of ML for the interpretation of data or the generation of case reports, higher evaluation criteria are needed, including the explainability of the model [49]. Furthermore, methods such as Shapley Additive exPlanations (SHAP) [114] for the interpretation of black-box models (e.g., XGBoost) or deep neural networks should be considered when appropriate (e.g. in the juridical context). Overall, at this stage, we believe ML should be used as a triage tool in routine use (e.g., retention time prediction, flagging atypical urine samples) rather than as a tool to make final decisions that could go into a report. Accordingly, defining concrete criteria for decision-making applications of ML in forensic toxicology is highly context-dependent, jurisdiction-specific, and subject to change as technologies evolve, requiring multidisciplinary expert input.

Historically, ML applications relied on data-handling and model-development approaches that were largely restricted to individuals with expertise in computer programming, as developing models required proficiency with complex libraries such as TensorFlow. More recent advances in abstraction and software design have since facilitated the creation of high-level application programming interfaces (APIs), thereby lowering technical barriers and enabling domain scientists without extensive technical expertise to construct models with relative ease. However, in an ideal scenario, methodological experts (e.g., biostatisticians, data scientists), who provide in-depth expertise in data handling and model development but may risk overlooking essential domain-specific aspects, team up with domain experts (e.g., forensic toxicologists), who contribute subject-matter knowledge but may lack the advanced data mining expertise. Hence, collaborative teams combining expertise from both fields are crucial to advance reliable and adequate research. Best practices need to be optimized to leverage the full potential of ML in forensic toxicology.

Taken together, the reviewed ML applications in forensic toxicology reveal several overarching limitations that currently constrain the quality and applicability of ML approaches. Across studies, dataset sizes are often limited, for example, by the restricted number of considered compounds (e.g., inclusions of only certain NPS) or by small sample sizes (e.g., limited number of animals; availability of samples from forensic case work). These constraints directly affect the robustness, generalizability, and validity of the resulting models and predictions. Another factor is that validation strategies are frequently restricted to internal cross-validation, in which datasets are split into training and test sets without using independent external validation data. Furthermore, systematic evaluation of overfitting is rarely reported. Other observed issues are that cross-instrument or cross-laboratory testing of algorithms is only scarcely addressed, limiting insight into their transferability. Collectively, these limitations highlight that, despite clear potential, most ML approaches in this field are presently best suited for supportive and exploratory tasks rather than for definitive forensic interpretation.

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Declaration of Competing Interest

None.

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